INCH-POUND
MIL-DTL-32235/4
30 March 2021

DETAIL SPECIFICATION SHEET

UNITIZED GROUP RATION-EXPRESS (UGR-E) HEATER MODULE: ASSEMBLED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32235.

CLASSIFICATION

Heater modules are of the following styles:

Style A - Three sub-units plus a Boil-In-Bag (BIB) module
Style B - Four sub-units

REQUIREMENTS

I. Heater module.

The heater module shall consist of a heater module box containing three or four heating trays, activation fluid units, heaters, polymeric food trays or Institutional Size Pouches (ISP) and the activation mechanisms/pull tabs. Sub-assemblies are allowed. When applicable, there shall also be a Boil-In-Bag (BIB) module. The components and sub-units shall be assembled and the activation mechanisms/pull tabs shall be connected to the three or four activation fluid units and to the three or four heaters. The sub-units and the BIB module (when applicable) shall be placed into the heater module box.

A. Heater.

The heater shall be constructed of materials that, when activated by a fluid, shall initiate and propagate an exothermic reaction suitable for use with food. This reaction shall generate adequate heat to completely cook the food when applicable, or heat the food to a safe food serving temperature. No toxic gas, liquid or solid by-products are desirable. If toxic by-products are produced, they shall be of the least severity and smallest amount possible while

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allowing for adequate heating and ensuring operator and consumer safety. When a low hydrogen generating heater is used, it shall generate less than 30 liters of hydrogen in 60 minutes. The heater material shall be evenly distributed and completely sealed within the scrim matrix of the heater to minimize the release of materials, and facilitate direct in-place activation of the heater materials. The heating rate shall be optimized to minimize the time required to heat the food, yet not cause excessive foaming or uncontrolled release of reaction by-products. The heater and barrier material shall not melt, deform or degrade during heating.

The heater is activated by the addition of a fluid that shall fully activate the heater material. The non-woven porous polymeric scrim shall be sealed and sized to accommodate proper fit and function of the heater module. Each heater (heater elements in a matrix) shall be correctly and legibly labeled in accordance with MIL-DTL-32235. In addition, the following statement shall be labeled on the barrier pouch and all boxes, as applicable:

FLAMELESS HEATER – DO NOT CONSUME

NOTE: Any hazard markings required for Occupational Safety and Health Administration (OSHA), U.S. Environmental Protection Agency (USEPA) and U.S. Department of Transportation (USDOT) compliance are to be labeled on the barrier pouch and all boxes, as applicable.

The heater shall be placed in the bottom of the heating tray, covered and hermetically sealed in place with a layer of barrier material to the bottom of the tray with a peelable seal. The barrier material shall be scored (laser or mechanical) and shall be attached to a pull strip to enable activation.

B. Activation fluid unit.

The activation fluid unit consisting of the pouch containing the activation fluid shall be made of material equivalent to Class 1 of MIL-PRF-131. Alternate activation fluid pouch materials and designs shall be permitted with approval from Combat Capabilities Development Command (DEVCOM) Soldier Center. The pouch shall be manufactured in accordance with the dimensions and design shown in Figure 1. Tolerances for the pouch dimensions shall be ± 1/8 inches. Sufficient length for the center strip and careful assembly is critical to ensuring that the pouch is not inadvertently torn open during assembly and subsequent transport and storage. In Figure 1, the solid lines shown at 1 inch off center at the base of the strip are cut lines. The 1 inch center strip section of the pouch shall be constructed with additional material for reinforcement. The center section of the pouch shall be scored (laser or mechanical) to provide easy tear properties without degrading the strength and barrier properties of the pouch. The pouch shall be filled with 1.5 percent saline (water and sodium chloride) solution, or as specified by the heater manufacturer with approval from Combat Capabilities Development Command (DEVCOM) Soldier Center. The volume of fluid in the pouch, when combined with the heater, shall be adequate to initiate and propagate

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the exothermic reaction. Each activation fluid unit shall be correctly and legibly labeled in accordance with MIL-DTL-32235. In addition, the following statement shall be labeled on the pouch:

ACTIVATION FLUID - DO NOT CONSUME

NOTE: Any hazard markings required for Occupational Safety and Health Administration (OSHA), U.S. Environmental Protection Agency (USEPA) and U.S. Department of Transportation (USDOT) compliance are to be labeled on the activation fluid unit and all boxes, as applicable.

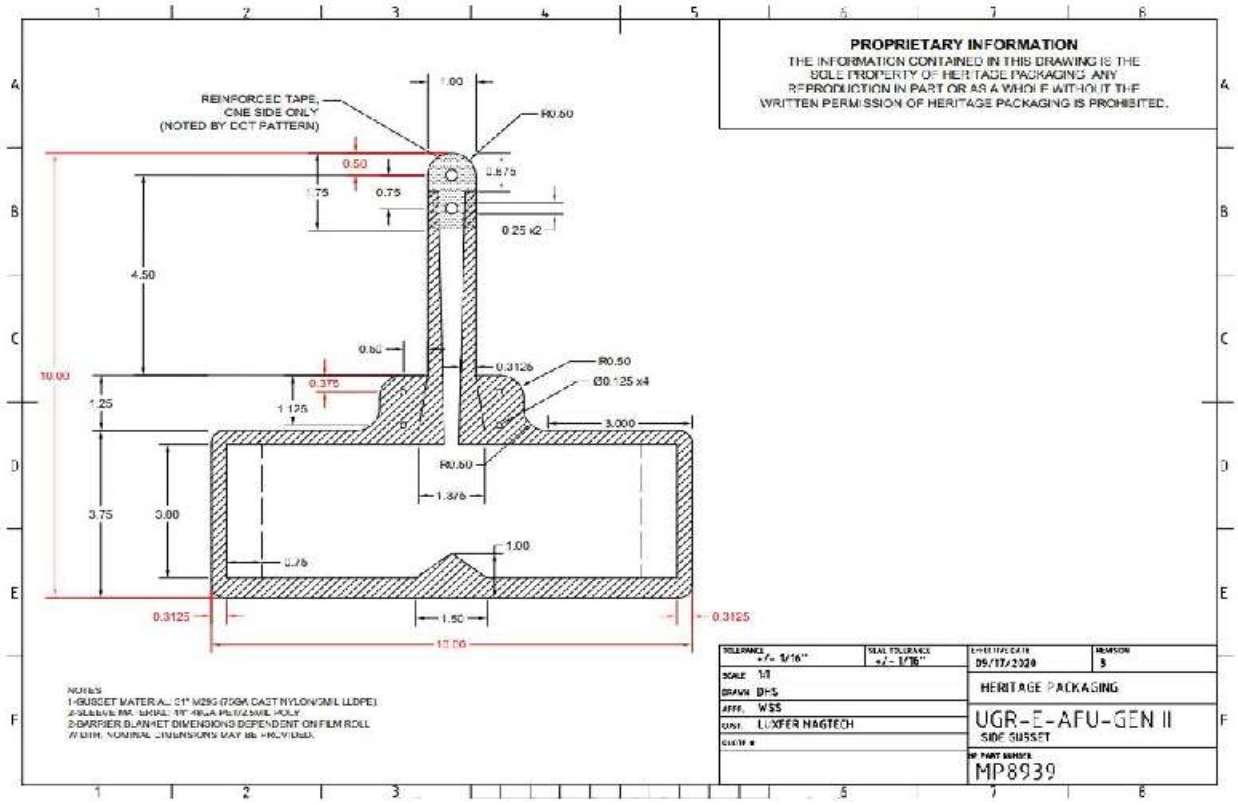


FIGURE 1. Activation fluid unit

C. Heating tray.

The heating tray shall be formed from solid or reinforced fiberboard having a minimum thickness of 0.040 inch. The fiberboard tray shall be constructed to provide a water/saline impermeable inner compartment suitable for containing the heating element and capable of fully supporting the filled/sealed polymeric food tray. The tray shall include a sealed-in heating element and the sealant layer shall consist of at least one aluminum foil

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(vapor barrier) layer and a high barrier sealable polymeric layer suitable for fully encapsulating the heater to provide extended shelf life to meet the requirements of this specification. The top heater sealant layer must include an extended center leader section that can be securely attached to an activator pull tab to facilitate a tear-away layer, or center section that is fully torn away when the activator tab is pulled for activation (in use). The tray shall be further configured to fully contain the activation fluid unit with a center leader that can be securely attached to a second activator pull tab located outside of the tray end nearest to the activation fluid unit to facilitate a tear-open center section that is fully torn away when the second activator tab is pulled for activation (in use). The heating tray shall consist of two compartments. The smaller compartment serves as a retaining reservoir for the activation fluid unit and includes two buttons to which the two short tabs of the activation fluid unit pouch are secured.

A fold-over flap shall be included on the heating tray to retain and protect the activation fluid unit. A slot shall be cut into the cover to allow the center strip of the activation fluid unit pouch to slide through, which later is attached to an activation mechanism/pull tab.

The larger compartment shall be configured to hold the heater, one polymeric food tray or ISP, and accommodate the activation fluid. The compartment shall be configured with a raised and rounded edge that supports the polymeric food tray securely above the heater.

Dimensions of the heating tray shall be as specified in Figure 2 and Heater Seal Lid Stock shall be as specified in Figure 3. The tolerance for the angle measurements shall be ± 1 degree. The tolerance for the linear measurements shall be ± 1/8 inch.

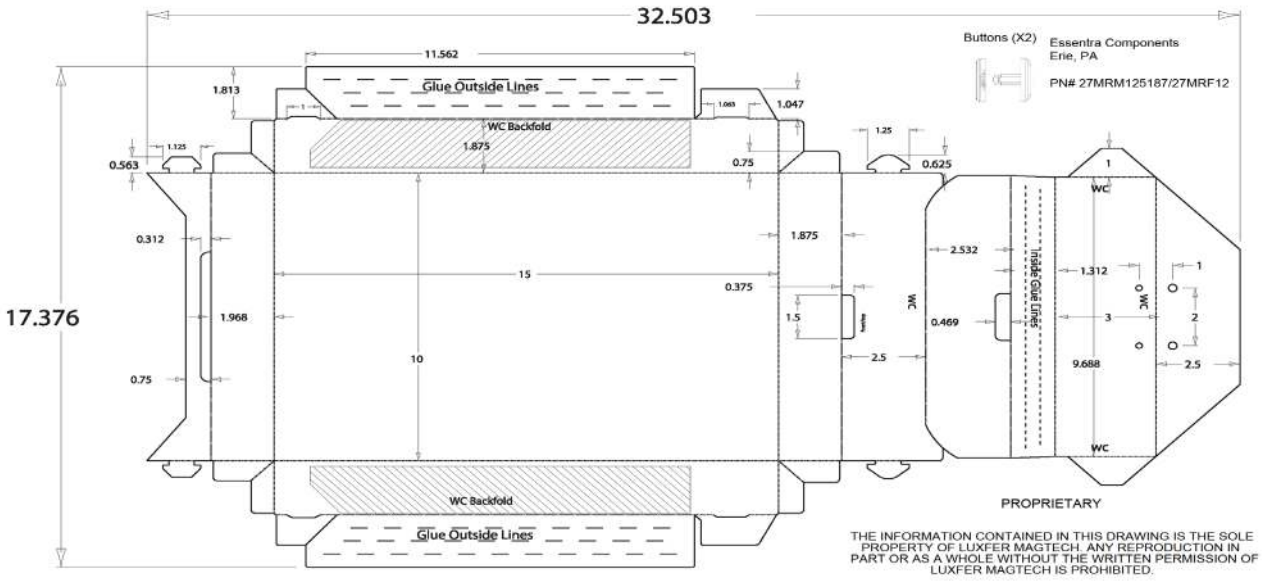


FIGURE 2. Heating tray

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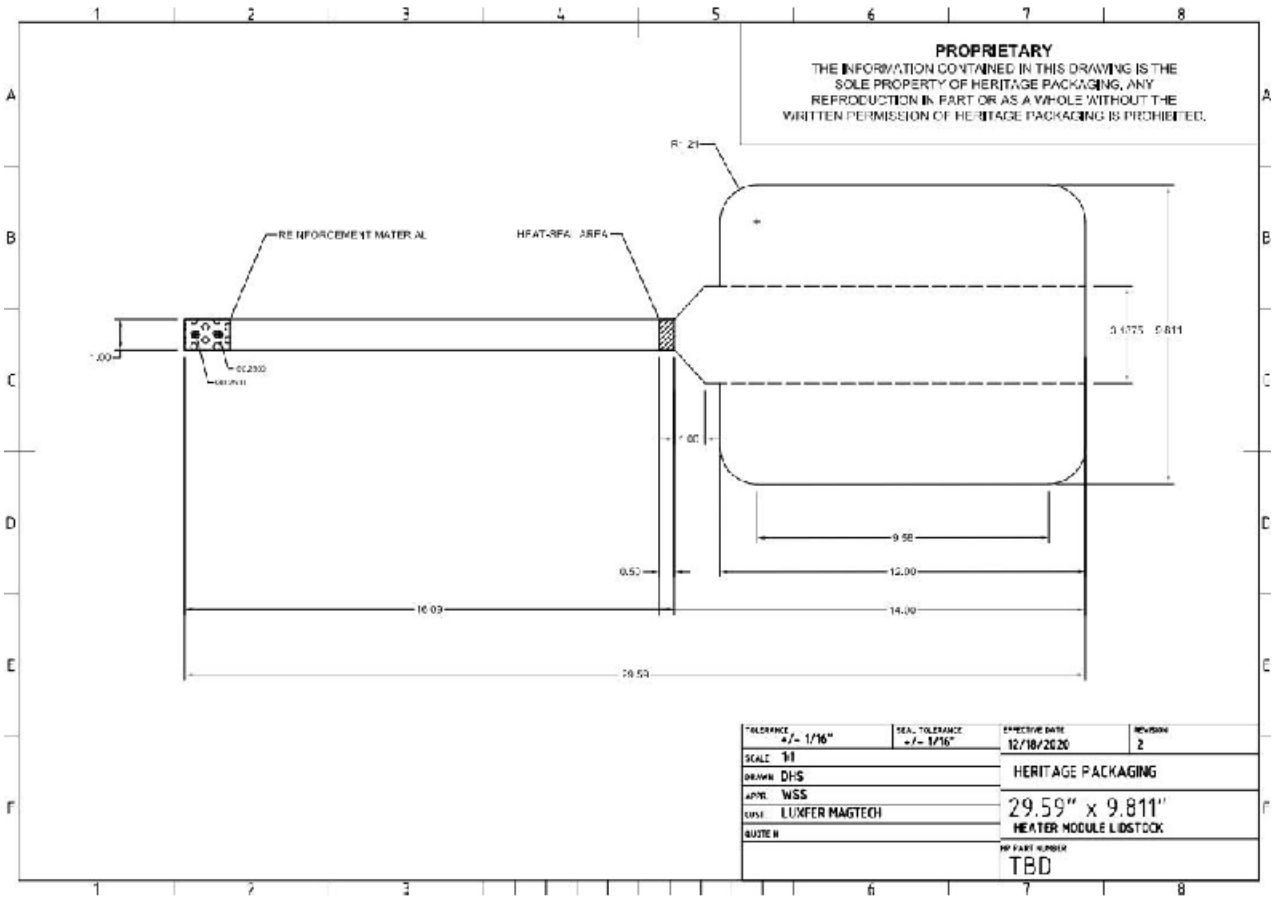


FIGURE 3. Heater Seal Lid Stock

D. Pull tabs.

The pull tabs shall be constructed of a blend of low to medium density polyethylene, copolymer polypropylene, or equivalent material and shall provide high strength characteristics under a wide range of environmental conditions. The material shall withstand temperatures ranging from -20°F to 160°F without fracture or failure. Dimensions of the pull tabs shall be as specified in Figures 4-1 through 4-3.

The pull tabs shall be configured with loading stations to support the three or four tray two-step activation method intended for the heater module. Each station shall be configured to retain the assembled activator strip and withstand a minimum pull force of 75 pounds. For a tab configured like the R16-HCEACTSTRIP-REV-C tab (Figures 4-1 through 4-5), the end of the center strip of the activating fluid unit shall be folded between the

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retaining holes and the tab inserted through the retaining station(s). The second pull tab shall be assembled to the heater pull strips in a similar configuration.

Upon completion of the heater module, the three or four center strips of the activation fluid units securely connect to a pull tab, and the three or four pull strips of the packaged heater securely connect to another pull tab. At the time of use, the operator of the heater module pulls the pull tabs to expose the heaters and to tear the activation fluid unit pouches which open and release the saline solution and activate the heaters.

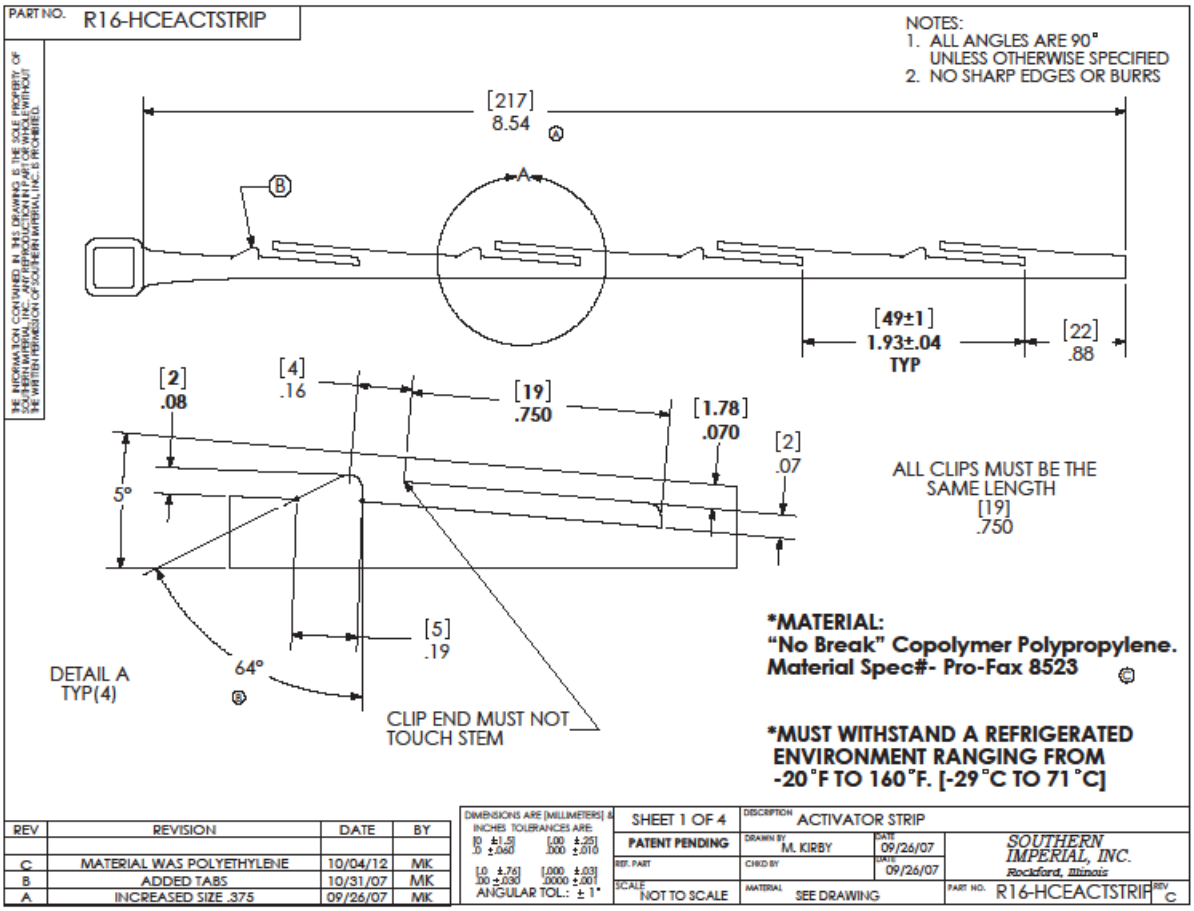
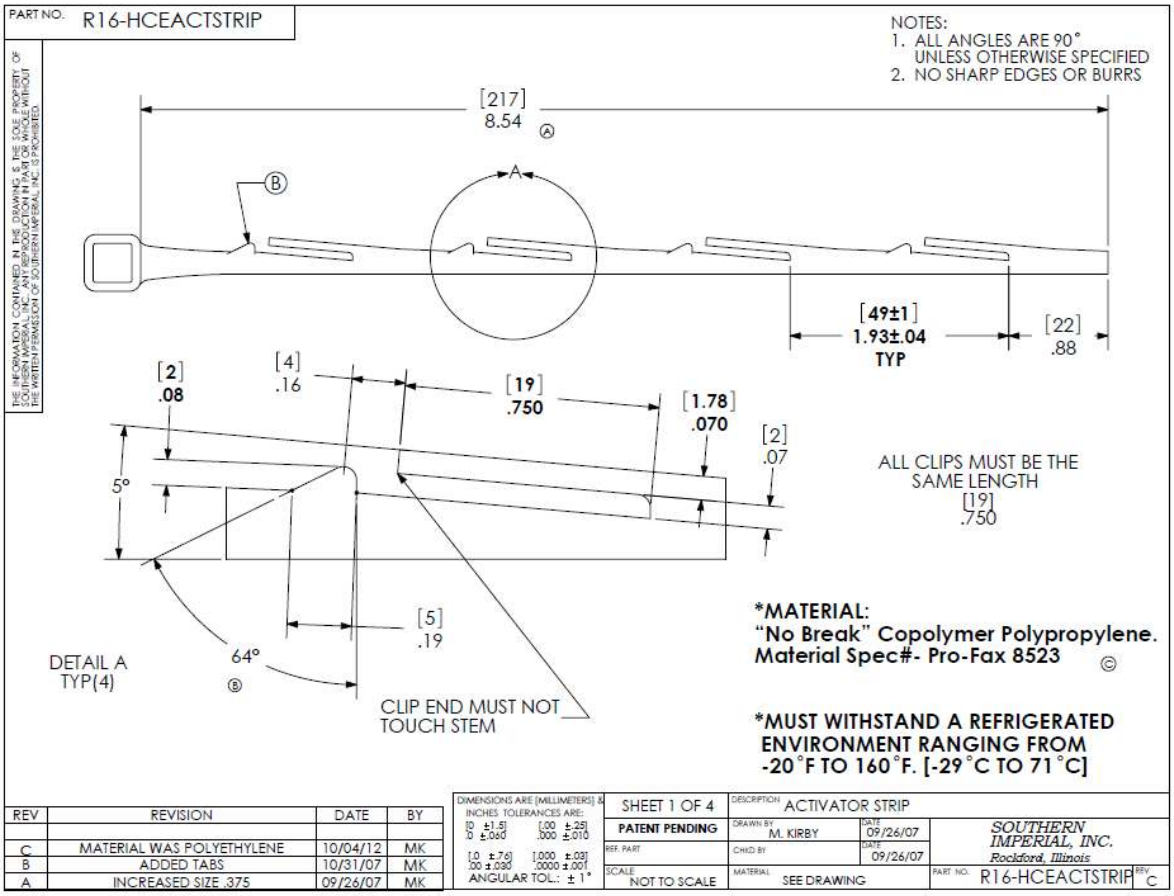


FIGURE 4-1. Pull tab, Center loading design

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REV	REVISION	DATE	BY
C	MATERIAL WAS POLYETHYLENE	10/04/12	MK
B	ADDED TABS	10/31/07	MK
A	INCREASED SIZE .375	09/26/07	MK

FIGURE 4-2. Pull tab, Center loading design

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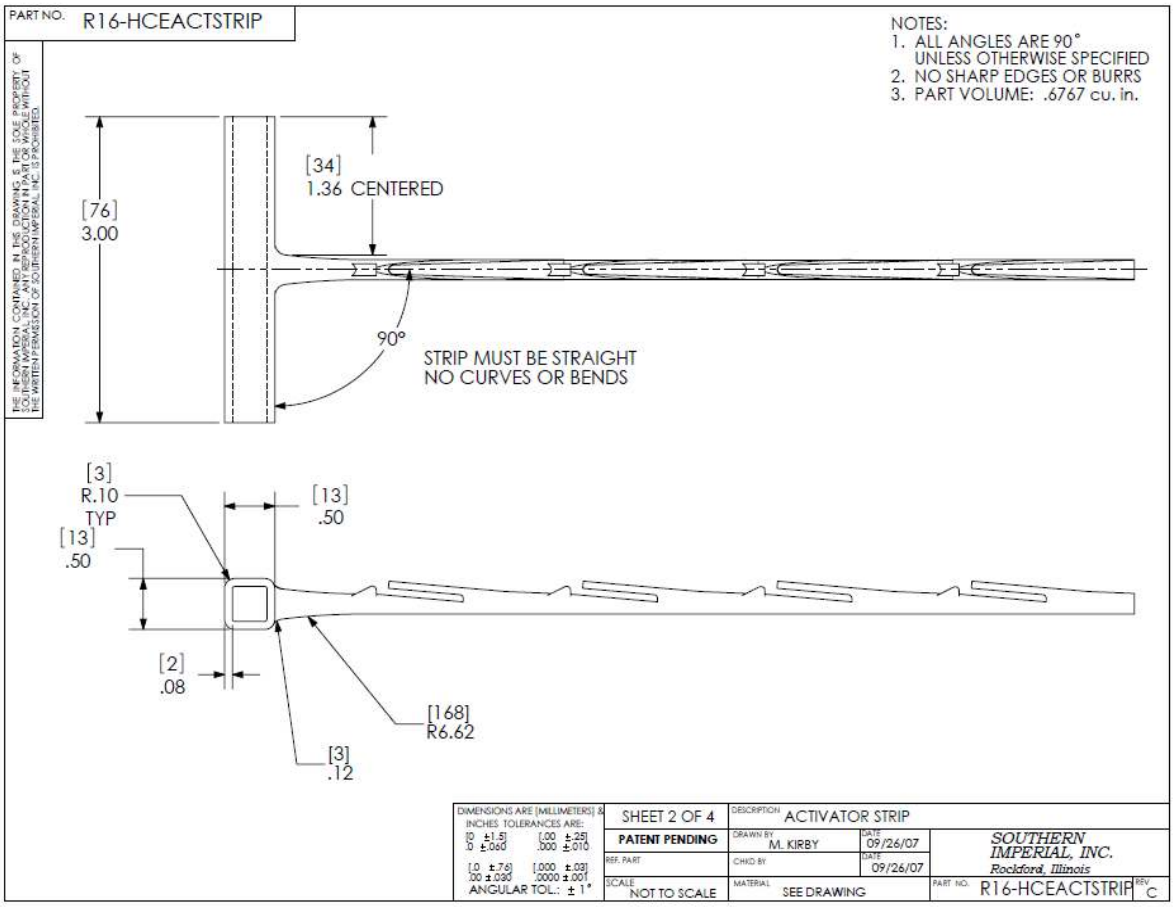


FIGURE 4-3. Pull tab, Center loading design

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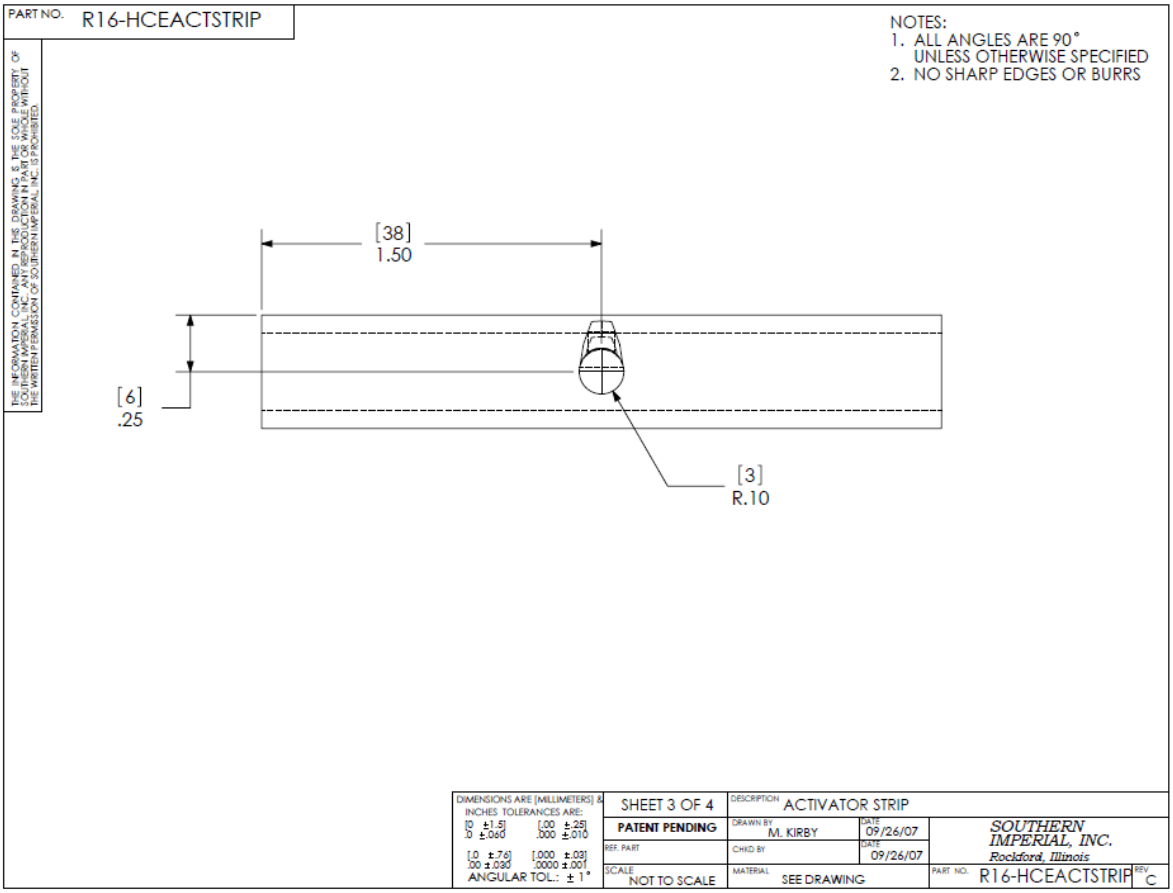


FIGURE 4-4. Pull tab, Center loading design

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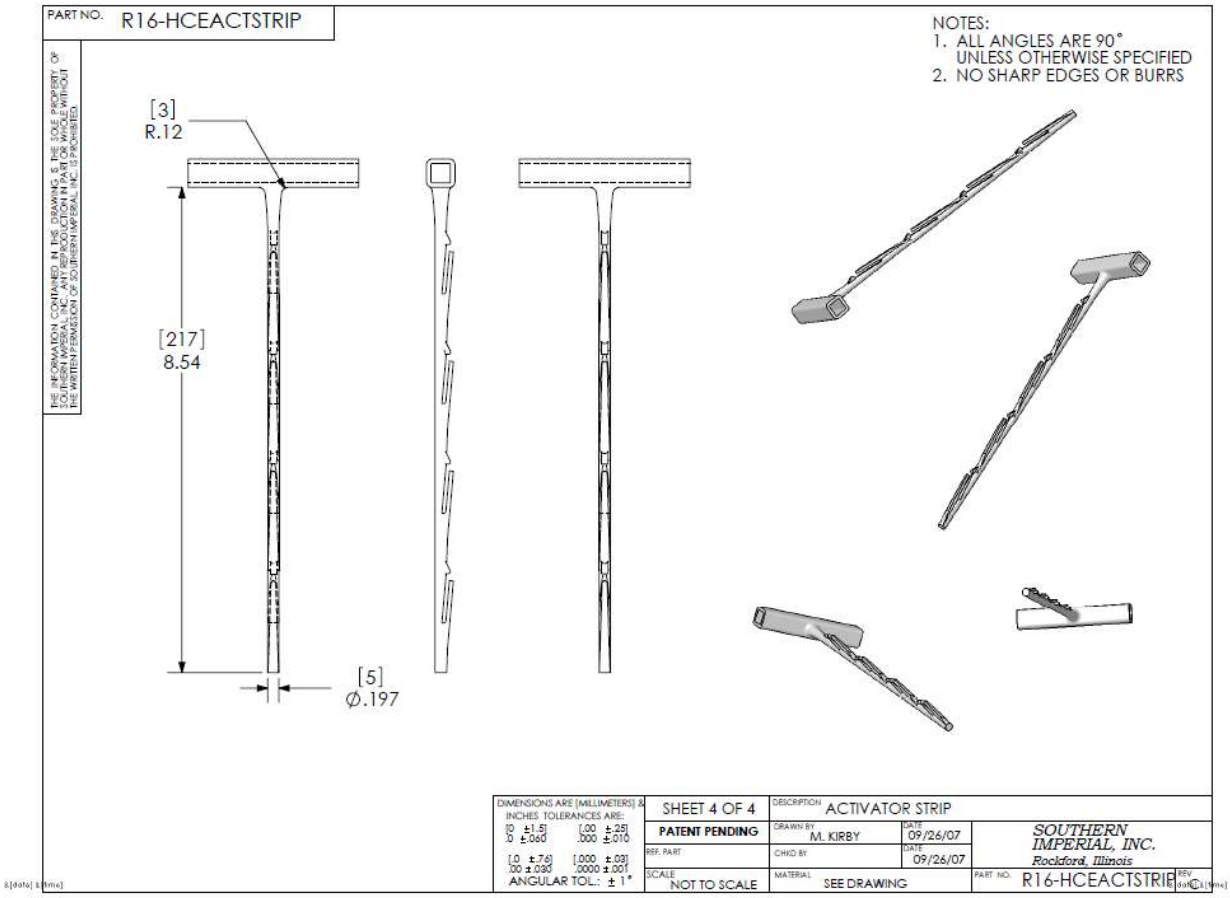


FIGURE 4-5. Pull tab, Center loading design

E. Assembly of heater module.

The components or sub-units shall be assembled. The heater module shall consist of either style A, a box containing three sub-units and a BIB module or style B containing four sub-units, with activation mechanisms/pull tabs. Each sub-unit shall consist of the heating tray with sealed-in heater, the activation fluid unit and the polymeric food tray or (ISP). Sub-assemblies of components may be used. The three or four sub-units shall be stacked and one activation mechanism/pull tab shall be connected to the three or four activation fluid unit center strips and the other activation mechanism/pull tab shall be connected to the three or four heater pull strips. The sub-units and the BIB module (when applicable) shall be placed into the heater module box. The entrée polymeric food tray shall be on the bottom and the

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dessert polymeric food tray shall be on the top. When there is a BIB module, it shall be placed on top of the three sub-units.

Corrugated fiberboard pads or cushioning shall be packed as necessary to fill excess headspace. The heater module box shall be closed and instruction sheets applied. Design and dimensions of the heater module box shall be as specified in Figure 5. Each heater module shall be correctly and legibly labeled in accordance with MIL-DTL-32235.

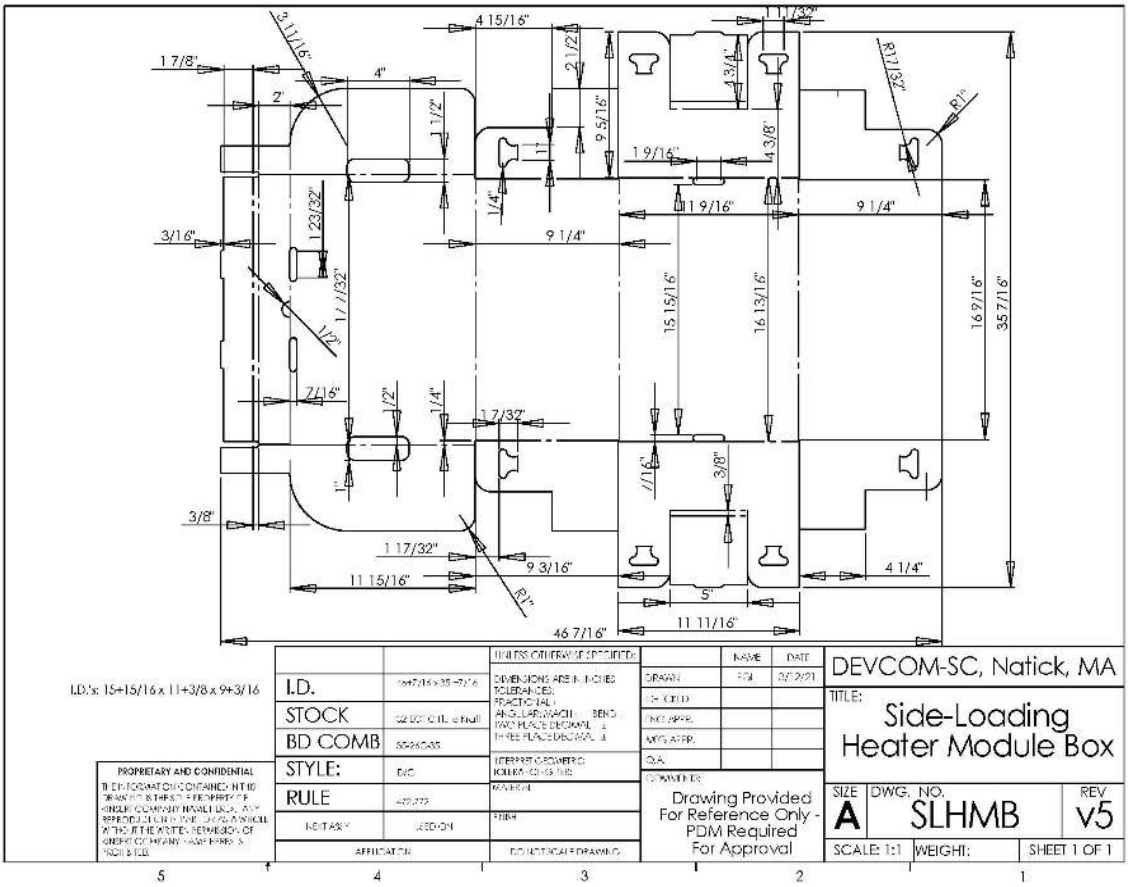


FIGURE 5. Heater module box

EXAMINATION AND TESTS

A. Heater module examination. The finished product shall be examined for conformance with the heater module examination requirements specified in MIL-DTL-32235 and this specification sheet. The heater module shall be examined for the defects listed in Table I.

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TABLE I. Heater module defects 1/ 2/ 3/ 4/ 5/ 6/ 7/

Category		Defect
Major	Minor	
101		Heater module not style specified.
102		Heater does not contain materials that will initiate and propagate an exothermic reaction.
103		Heater causes excessive foaming or uncontrolled release of reaction by-products.
104		Heater not hermetically sealed in tray with barrier material.
105		Heater module does not generate adequate heat to completely cook the food when applicable, or does not heat the food to a safe serving temperature. 8/
106		Center strip of activation fluid units not reinforced.
107		Score lines on activation fluid units missing or damaged.
	201	Score line on heater barrier material not provided.
	202	Heater barrier material not connected to pull strip.
	203	Heater pull strip not attached to activation mechanism/pull tab.
108		Activation mechanism/pull tab missing or damaged.
	204	Activation fluid unit pull strip not attached to activation mechanism/pull tab.
	205	Activation mechanism/pull tab not assembled properly.
109		Fold-over flap on heating tray to retain the activation fluid unit missing or damaged.
110		Heating tray design or dimensions not correct.
	206	Polymeric trays or ISPs or BIBs (if applicable) of food not placed in module in correct order.

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TABLE I. Heater module defects 1/ 2/ 3/ 4/ 5/ 6/ 7/ - Continued

Category		Defect
Major	Minor	
111		Excess headspace in module allowing movement of components.
	207	Low hydrogen generating heater generates more than 30 liters of hydrogen in 60 minutes.

1/ Heater material construction shall be verified by Certificate of Conformance (CoC).

2/ Material in accordance with MIL-PRF-131 for the activation fluid unit shall be verified by CoC.

3/ The activation fluid solution shall be identified and verified by CoC.

4/ The use of 0.090 inch high density polyethylene for the heating tray shall be verified by CoC.

5/ The pull tab material shall be verified by CoC.

6/ The pull tab pull strength shall be verified by CoC.

7/ The low hydrogen generating heater shall be verified by CoC.

8/ BIB Eggs must be cooked to an internal temperature of 145˚F (63˚C) for fifteen seconds. Polymeric food trays must be reheated to an internal temperature of at least 135˚F (57˚C).

B. Test methods.

I. Single heater capacity test.

The objective of the heating capacity test is to verify that a single heater increases the temperature of the water in the polymeric tray by 100°F (from 40°F to 140°F) in 45 minutes or less. In this test, one sub-unit (polymeric tray of water in the heating tray with the heater and the activation fluid unit) is tested. The following procedures are should be followed:

1. Pre-condition 96 oz. water-filled test tray to a temperature range of 35°F to 40°F.

2. Align matching bi-metallic (copper-constantan) pegs of C-10 Locking Connector to bi-metallic holes in C-9 Locking Receptacle. Thread C-10 Locking Receptacle and C-9 together until seated.

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3. Connect Thermocouple wire installed on Locking Receptacle to data acquisition or computer terminal calibrated to the copper-constantan thermocouple.

4. Assemble heater and food tray within the heating tray. Add activation fluid or use activator pouch to activate heater.

5. The test shall be conducted at an ambient temperature of 72°F ± 2°F in an explosion-proof exhaust fume hood or sufficiently ventilated environment, away from open flame or potential ignition sources.

6. Place sub-unit into a representative heater module box with 10-3/4 by 16-1/4 inch corrugated insert placed over the tray (weather grade corrugated or plastic materials may be used for repeated testing). Activate the sub-unit.

7. Record temperature for at least 45 minutes at approximately 1 minute or less intervals.

PART IDENTIFIERS AND SOURCES OF SUPPLY

1. Heater. The heater is available from:

Luxfer Magtech Incorporated
2940 Highland Avenue
Cincinnati, OH 45212
(800) 503-4483

2. Heater barrier pouch. The barrier pouch material is available from:

Winter-Wolff International
131 Jericho Turnpike
Jericho, NY 11753
(516) 997-3300

3. Activation fluid unit. The activator pouch is identified as Part # UGR-E-AFU-GEN II. The material equivalent to Class 1 of MIL-PRF-131 for the construction of the activation fluid unit pouch is available from:

Cadillac Products Packaging Company
5800 Crooks Road
Suite 200
Troy, Michigan 48098-2830
(800) 837-0055

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The filled and sealed activation fluid units are available from:

Heritage Packaging
625 Fishers Run
Victor, NY 14564
(585) 742-3310

4. Heating tray. The heating tray is available from:

Luxfer Magtech Incorporated
2940 Highland Avenue
Cincinnati, OH 45212
(800) 503-4483

5. Pull tab. The pull tab is identified as Part # R16-HCEACTSTRIP-REV-C. The pull tab is available from:

Siffron Incorporated
8181 Darrow Road
Twinsburg, OH 44087-2303.
1-800-422-2547

6. Thermocoupled polymeric trays. Filled thermocoupled polymeric trays or instructions on how to construct them are available from:

Combat Capabilities Development Command (DEVCOM) Soldier Center
FCDD-SCD-SCR
10 General Greene Avenue
Natick, MA 01760-5000
(508) 206-3410

7. Heating Tray with Sealed in Heater or Heating Tray with Sealed in Heater and Activator Pouch installed. The heater sealed into a heating tray or heater sealed into a heating tray with activator pouch installed is available from:

Luxfer Magtech Incorporated
2940 Highland Avenue
Cincinnati, OH 45212
(800) 503-4483

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8. Assembly and instruction sheets. The following assembly and operating instruction sheets are attached:

FIGURE 6. UGR-E Operating Instructions, for Heater with Hydrogen Generating Warning

FIGURE 7. UGR-E Operating Instructions, for Heater with Low Hydrogen Generating Warning

FIGURE 8. UGR-E BIB Operating Instructions, for Heater with Hydrogen Generating Warning

FIGURE 9. UGR-E BIB Operating Instructions, for Heater with Low Hydrogen Generating Warning

FIGURE 10. Instructions for UGR-E Eggs, for Heater with Hydrogen Generating Warning

FIGURE 11. Instructions for UGR-E Eggs, for Heater with Low Hydrogen Generating Warning

FIGURE 12-1. Assembly Instructions for UGR-E Heater Module

FIGURE 12-2. Assembly Instructions for UGR-E Heater Module - Continued

Assembly and instruction sheets in color are available electronically from:

Combat Capabilities Development Command (DEVCOM) Soldier Center
FCDD-SCD-SCR
10 General Greene Avenue
Natick, MA 01760-5000
(508) 206-3410

REFERENCES

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32235	-	Unitized Group Ration – Express (UGR-E) Heater Module, General Specification for
MIL-PRF-131	-	Barrier Materials, Watervaporproof, Greaseproof, Flexible, Heat-Sealable

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GOVERNMENT PUBLICATIONS

U.S. DEPARTMENT OF LABOR

Occupational Safety and Health Administration (29 CFR Part 1910, Subpart H)

(Copies of this document are available online at http://www.osha.gov or U.S. Department of Labor Occupational Safety & Health Administration, 200 Constitution Avenue, N.W., Room Number N3626, Washington, D.C. 20210.

U.S. DEPARTMENT OF TRANSPORTATION

Research and Special Programs Administration (49 CFR Parts 171-180)

(Copies of this document are available online at https://www.fmcsa.dot.gov or Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. ENVIRONMENTAL PROTECTION AGENCY

Resource Conservation and Recovery Act (40 CFR, Parts 239-282)

(Copies of this document are available online at http://www.epa.gov or from the Superintendent of Documents, ATTN: New Orders, P.O. Box 371954, Pittsburgh, PA 15250-7954.)

C. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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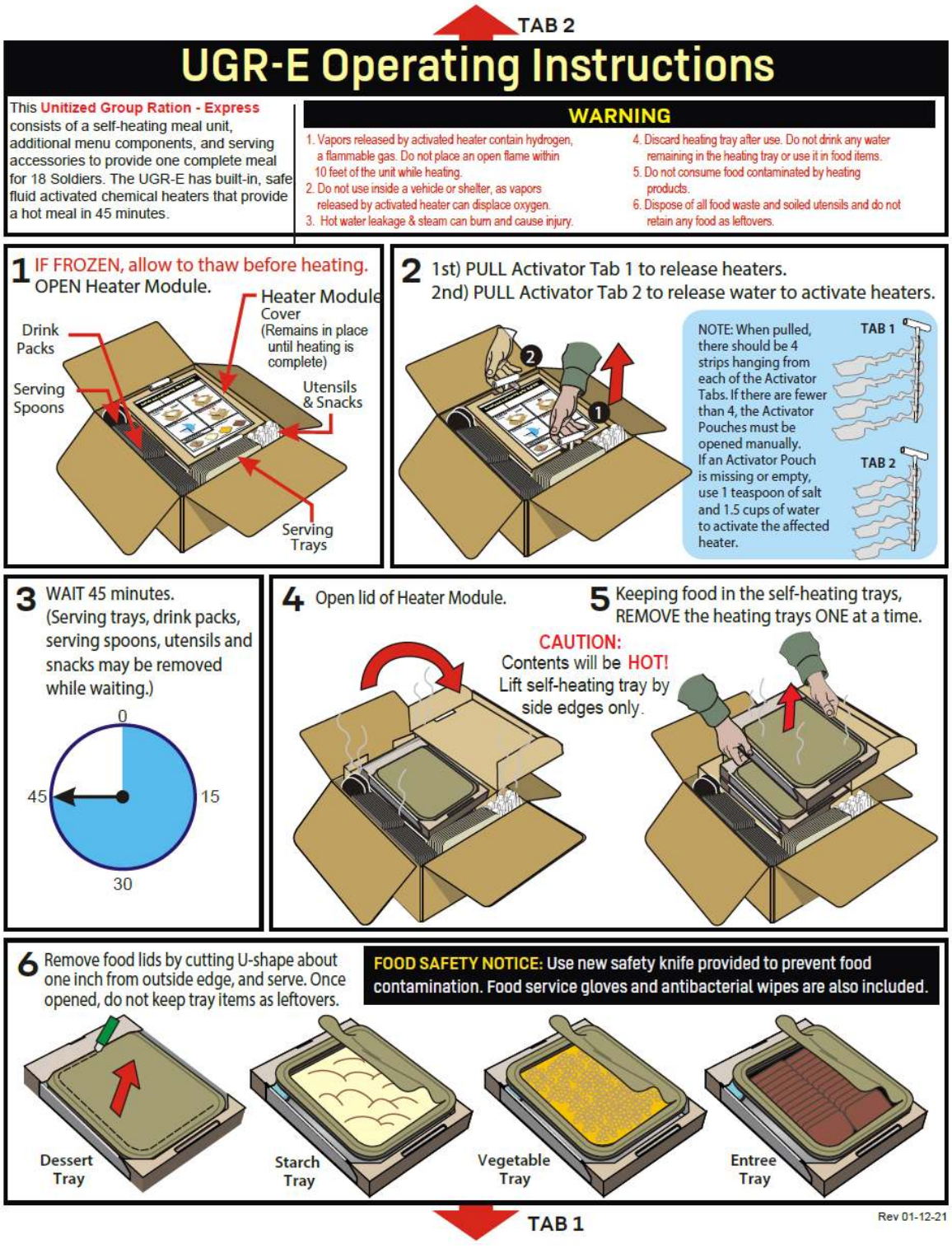


FIGURE 6. UGR-E Operating Instructions, for Heater with Hydrogen Generating Warning

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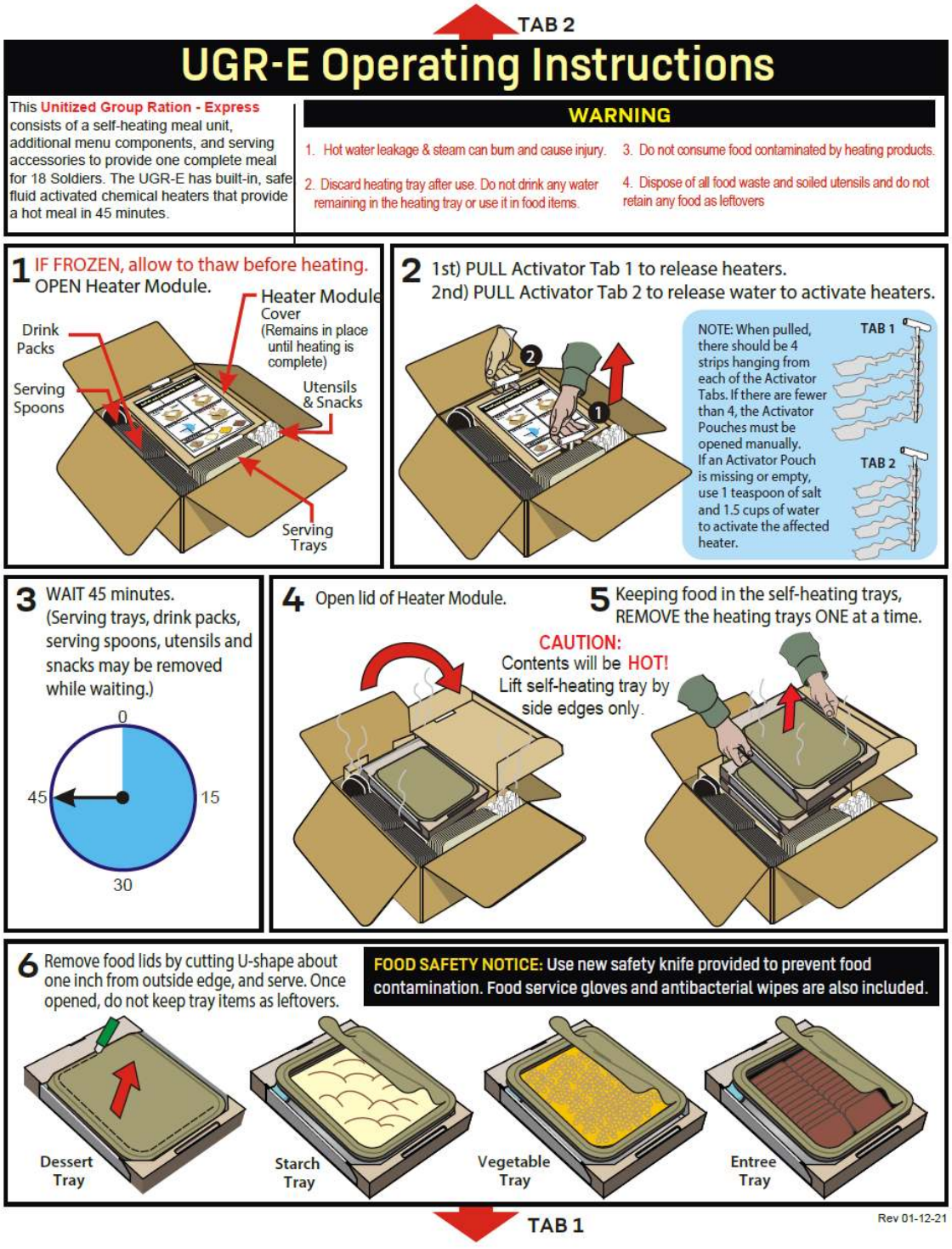


FIGURE 7. UGR-E Operating Instructions, for Heater with Low Hydrogen Generating Warning

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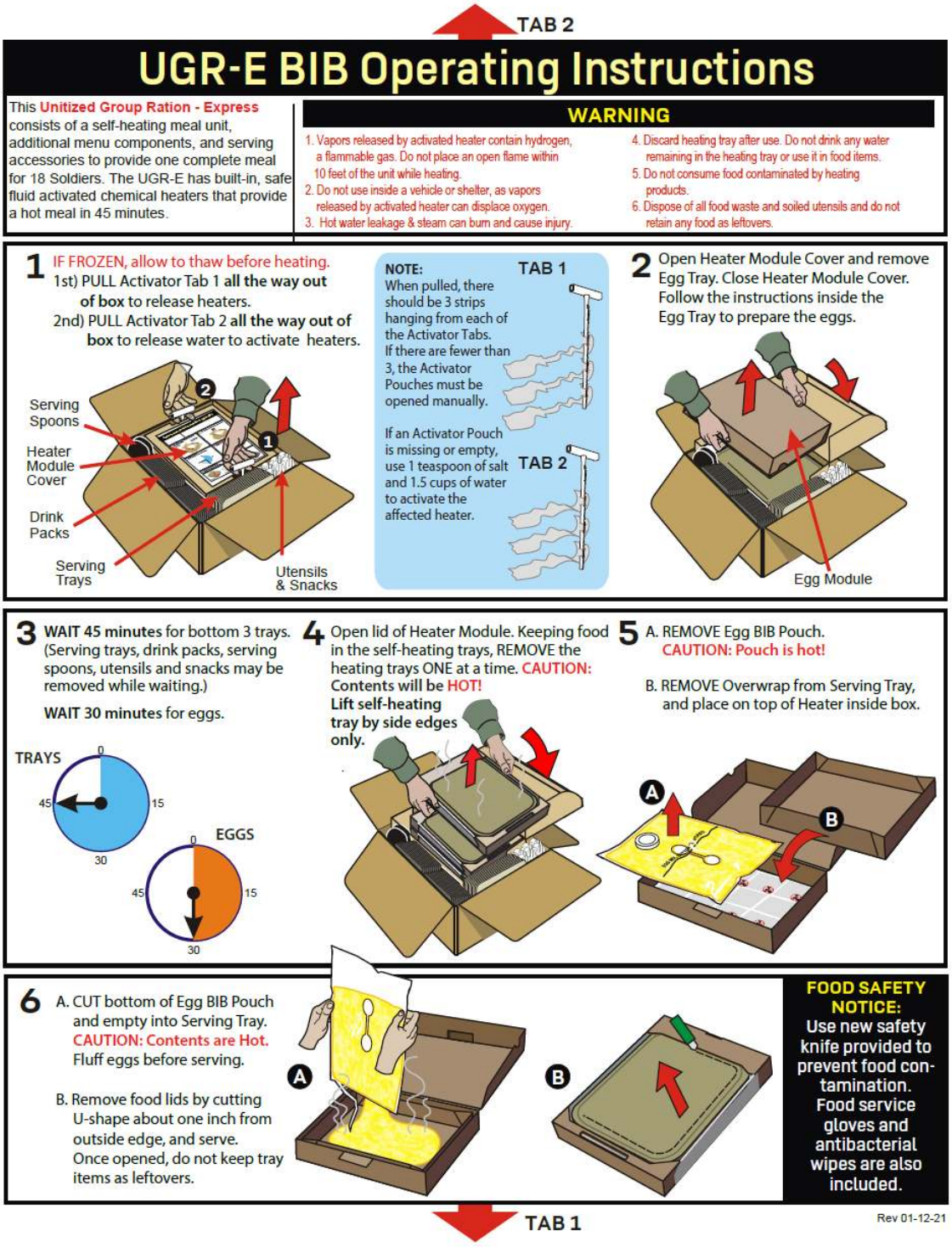


FIGURE 8. UGR-E BIB Operating Instructions, for Heater with Hydrogen Generating Warning

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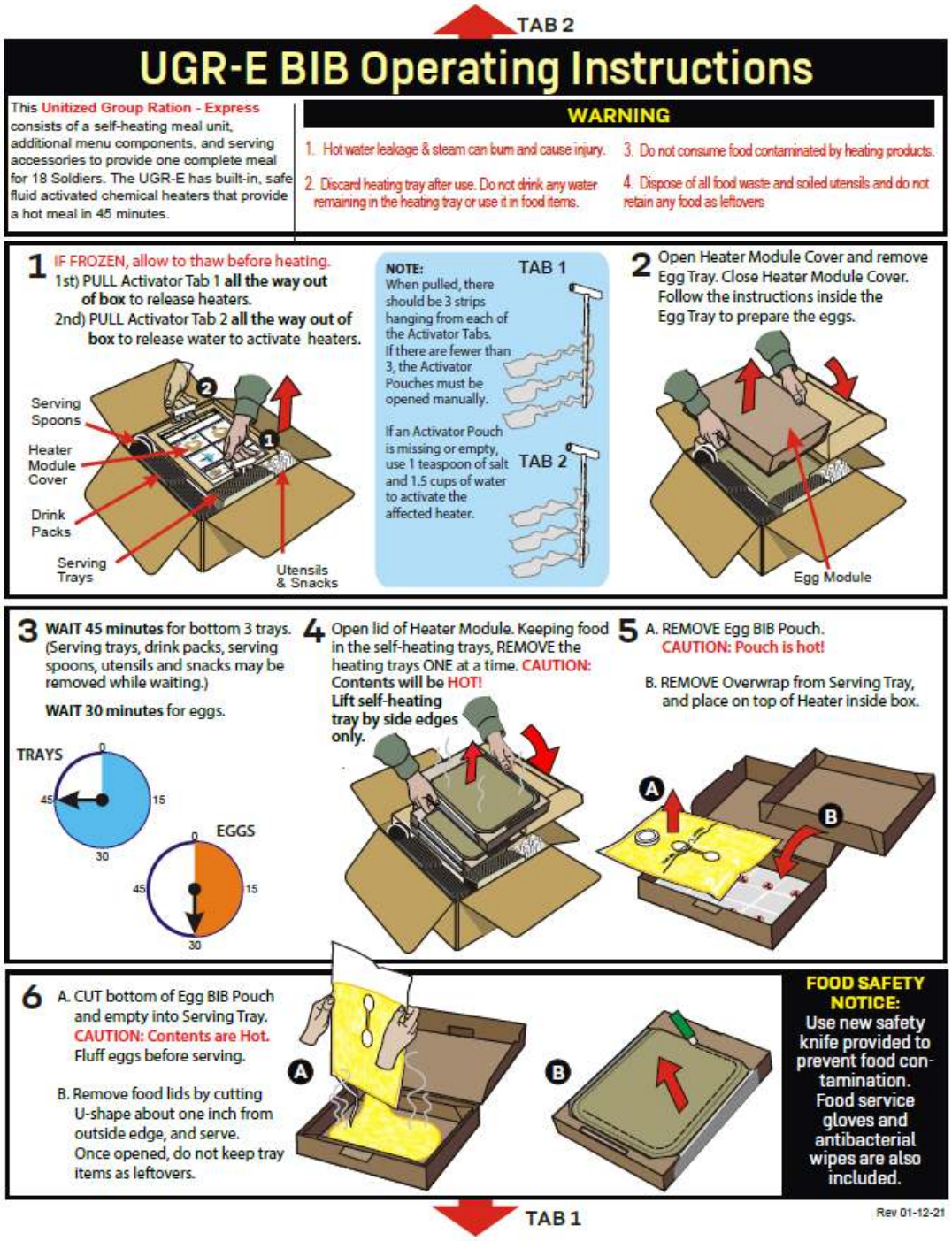


FIGURE 9. UGR-E BIB Operating Instructions, for Heater with Low Hydrogen Generating Warning

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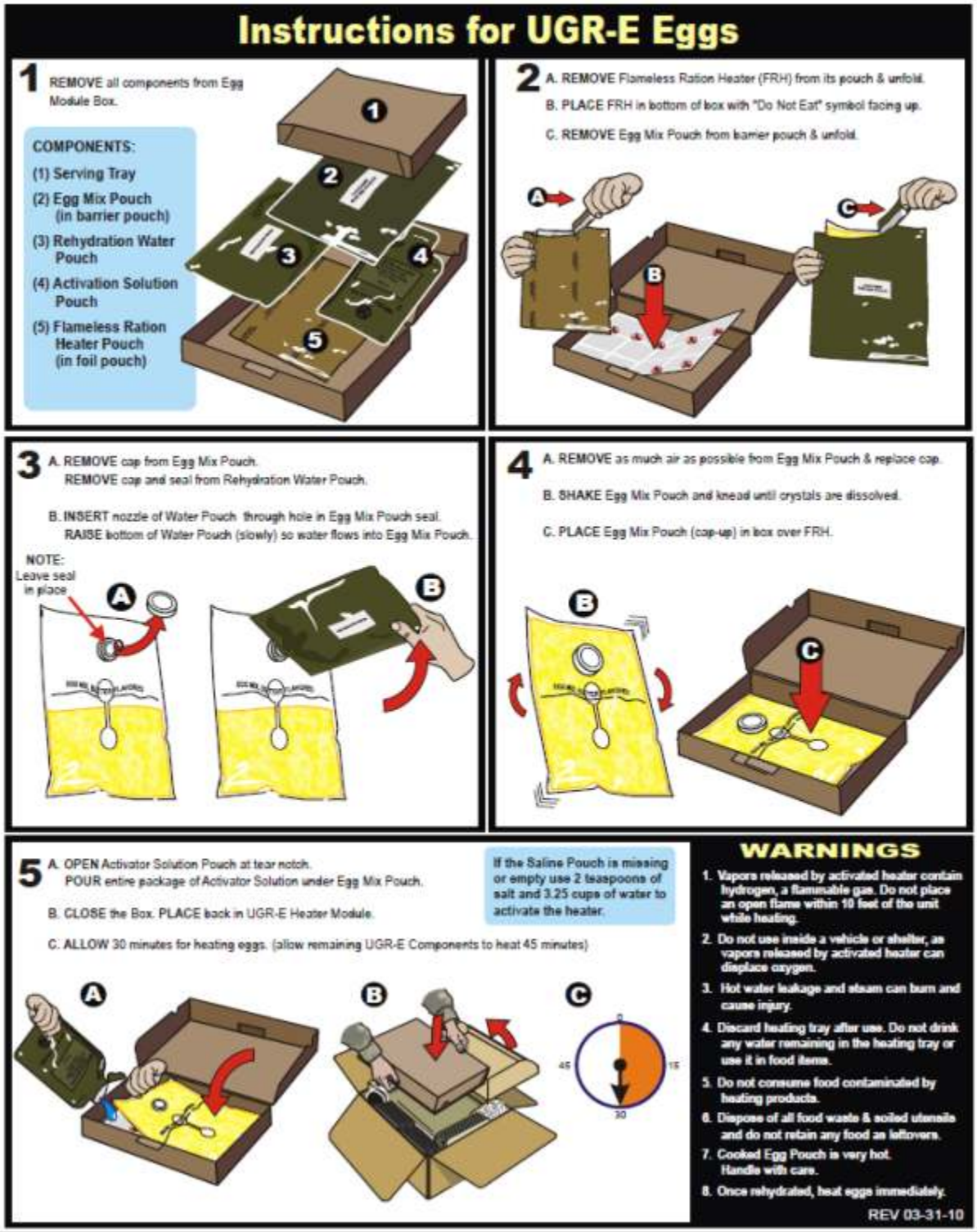


FIGURE 10. Instructions for UGR-E Eggs, for Heater with Hydrogen Generating Warning

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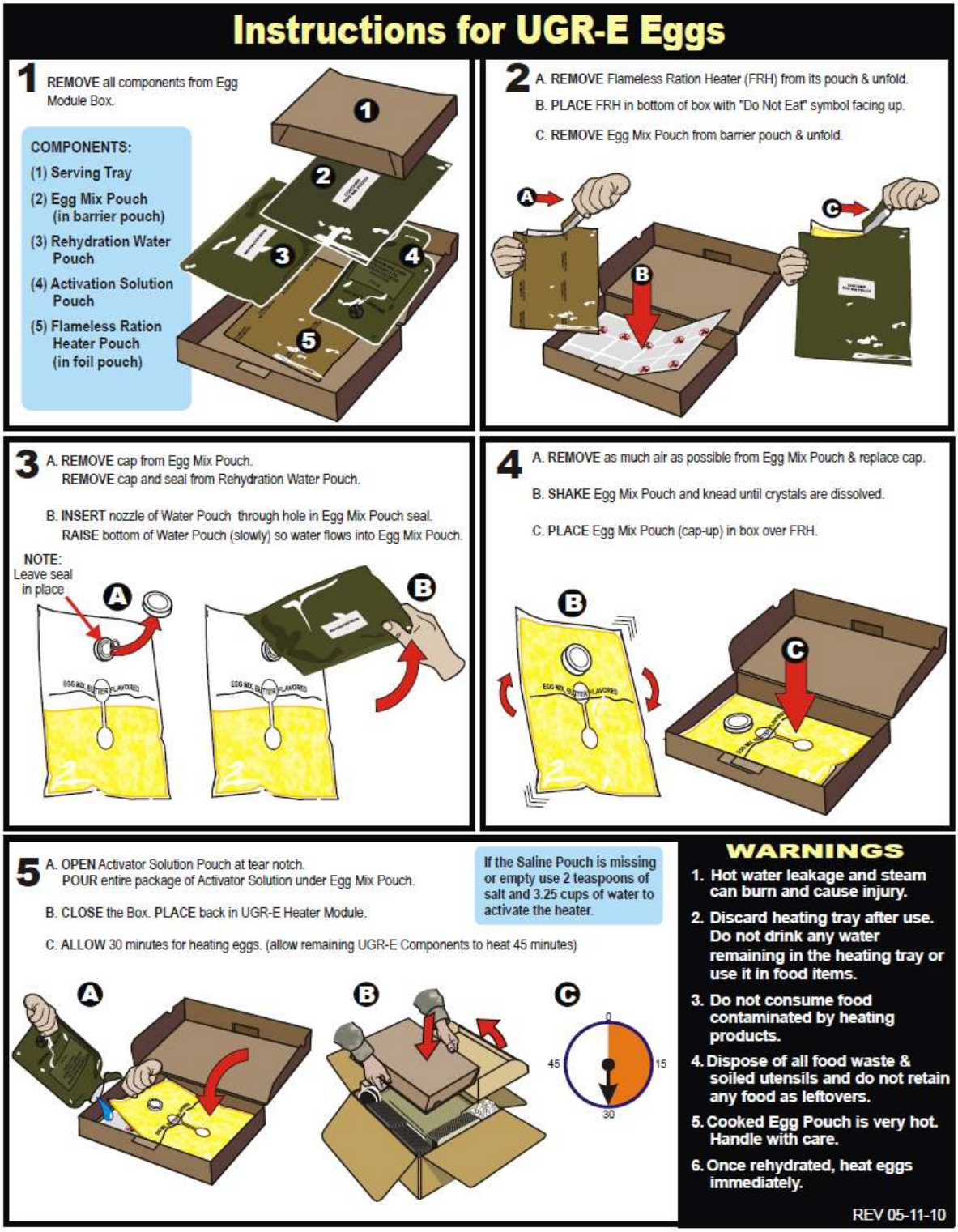


FIGURE 11. Instructions for UGR-E Eggs, for Heater with Low Hydrogen Generating Warning

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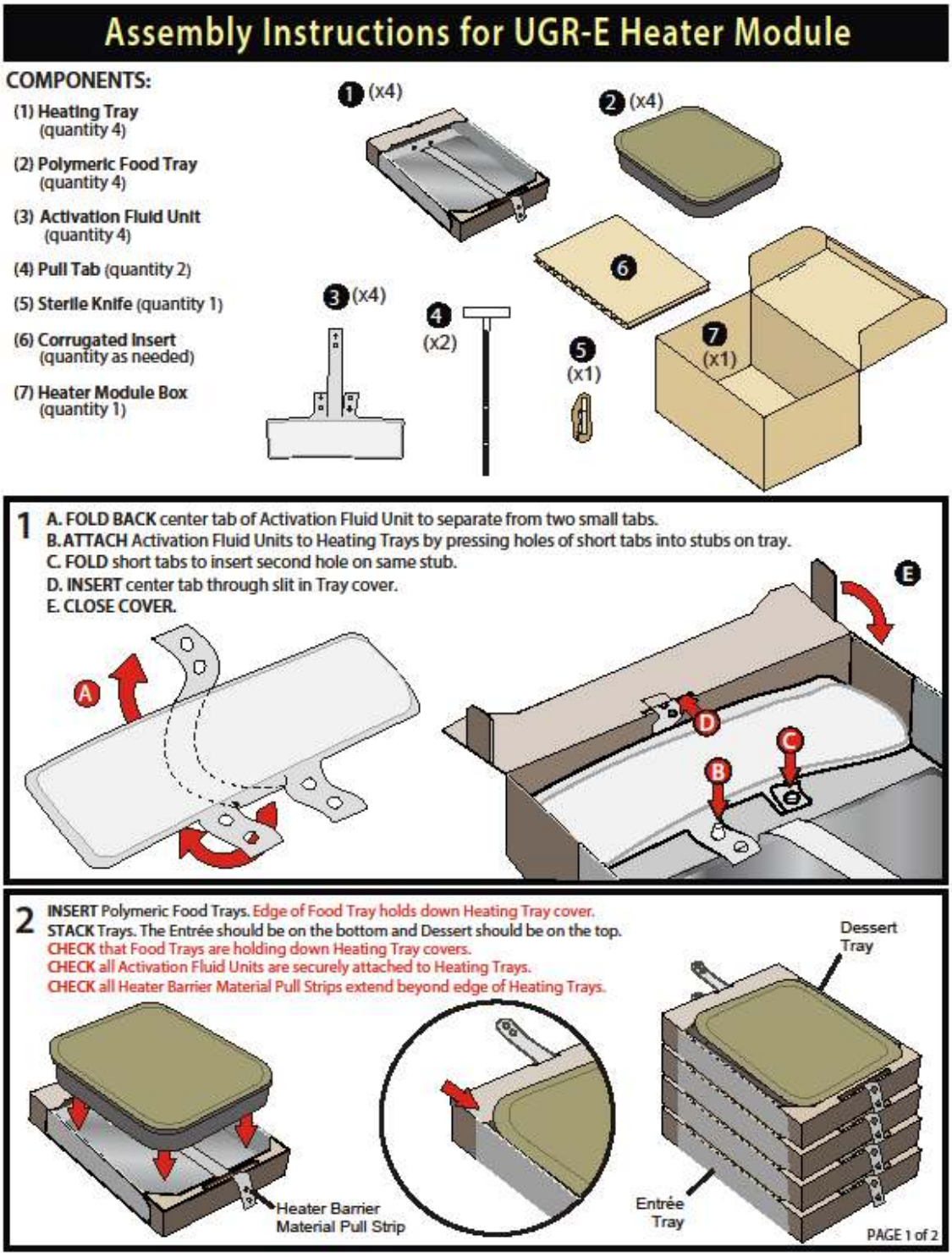


FIGURE 12-1. Assembly Instructions for UGR-E Heater Module

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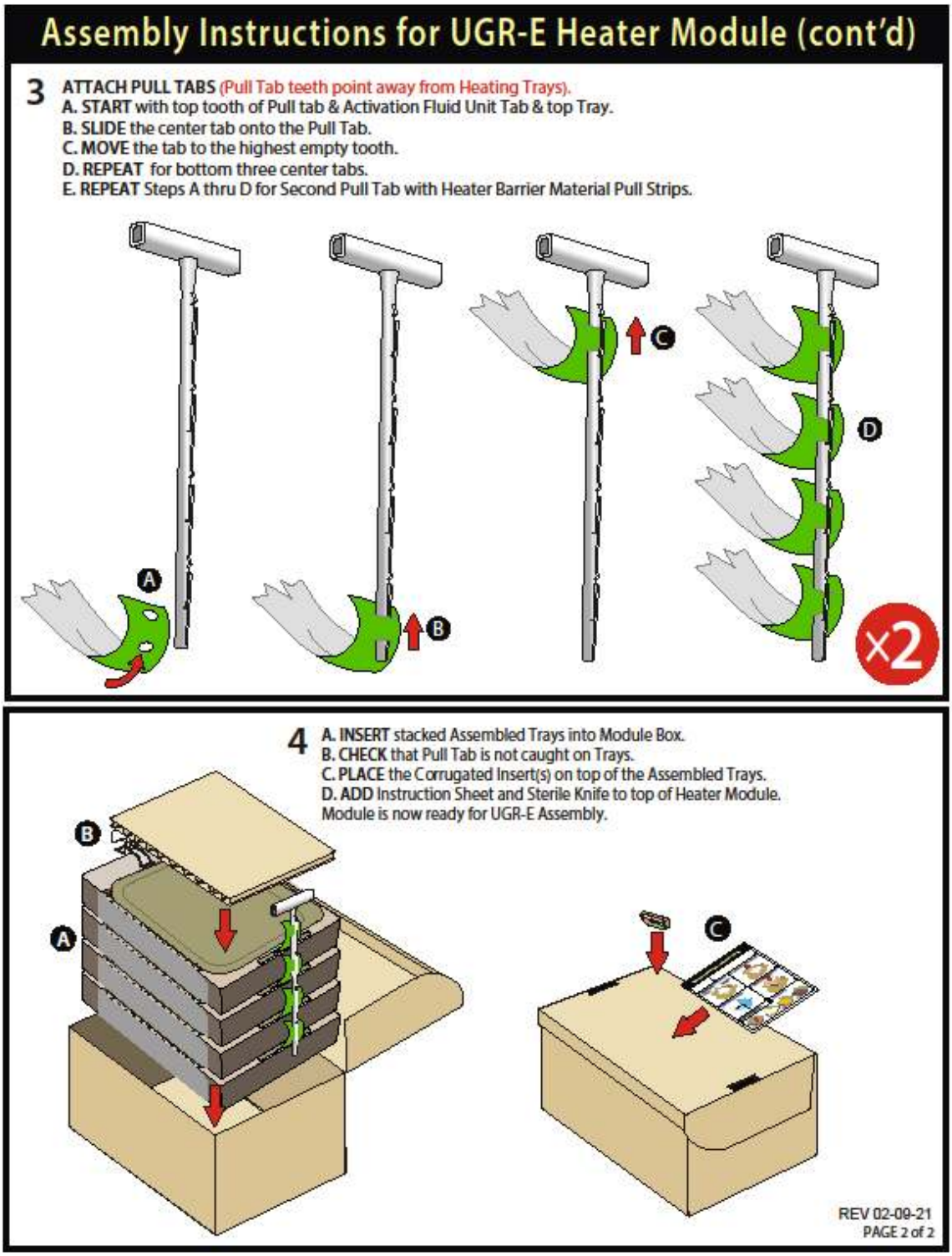


FIGURE 12-2. Assembly Instructions for UGR-E Heater Module – Continued

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Check the source to verify that this is the current version before use.

Custodians:
Army – GL
Navy – SA
Air Force – 35

Preparing activity:
Army – GL
(Project 8970-2021-006)

Review Activities:
Army – MD, QM
Navy – MC
DLA – SS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:26Z
Check the source to verify that this is the current version before use.